Fiscal Year:	FY 2013	**Task Last Updated:**	FY 05/20/2013
PI Name:	Rumschitzki, David Ph.D.		
Project Title:	How important is the contact angle in nucleate boiling heat transfer?		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
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Comments:			
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Key Personnel Changes/Previous PI:	Continuing Postdoc: Lin Zheng, PhD. ; Continuing Undergraduate candidate: Mr. Omar Duran (minority student), 4th year BS candidate in Chemical Engineering.		
COI Name (Institution):	Lee, Taehun (City College of the City University of New York) Koplik, Joel (City College of the City University of New York)		
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Task Description:	High latent heats allow evaporating small amounts of fluid to cool hot surfaces very efficiently. Boiling is thus attractive for cooling nuclear reactors, computer chips, hot surfaces in space (where volume is at a premium), etc,. provided the hot surface remains wet, which can be an issue in terrestrial and certainly is in microgravity. Dryout can be catastrophic. Despite decades of study, competing mechanisms for boiling heat transfer exist, including fluid mixing and liquid microlayer (under a portion of the vapor bubble) evaporation. We have carried out preliminary quasistatic heat conduction calculations of a single vapor bubble on the surface of a thin solid layer covered by a liquid of finite thickness. The solid sits on a heater at fixed temperature above the fluid's saturation temperature. The system is axisymmetric with periodic boundary conditions at its lateral edge to mimic bubble-bubble interactions. These conduction-only results match the pronounced heat transfer jump upon (surface) vapor bubble nucleation in all standard boiling curves and provide scalings of the heat flux with many key physical parameters that agree with historic correlations. Its simple mechanism may be transformative: The bubble's 3-phase contact line (CL) locally cools the superheated solid surface from an otherwise superheated value to the fluid saturation temperature. The contact line thus acts as a heat magnet from the solid for, e.g., microlayer vaporization. Critical questions include: How does the heat transfer vary as bubbles grow, deform, and detach? Does a detaching bubble leave behind a vapor residue/CL, generating a new bubble to draw heat? Does it detach cleanly? Or, does cooler fluid that replaces the rising bubble condense the residue bubble? A dynamic calculation addressing these questions needs CL motion with bubble growth as an input. Residue bubble quenching involves convection/convective transfer. We shall: 1. Create a molecular-sized model of our solid-liquid-vapor bubble system and let interparticle physics via molecular dynamics (MD) determine CL motion. We calculate bubble growth and heat transfer for a nanobubble directly from the MD results or input its CL motion into our quasistatic heat conduction model. This MD will begin with zero gravity and later introduce a pseudogravity so as to mimic a real bubble's surface tension-bouyancy balance (Bond number). 2. Formulate the fluid mechanics /heat transfer /phase change as a coarse-grained lattice-Boltzmann (LB) problem, which treats the liquid, its vapor and their mutual interface (including the CL) as a single fluid with a diffuse interface. One specifies the fluid-solid interaction energy; CL motion is part of the solution. This avoids the need to explicitly track the vapor-liquid interface and apply interfacial conditions there. In the presence of gravity, we use LB to follow deforming bubbles towards detachment for different system parameters, examine if a vapor residue remains on the surface after detachment and, if so, if convection quenches it. In gravity's absence we shall examine if the circulation provided by the expanding bubble's fluid displacement can help remove a bubble from the surface. We shall introduce intermittent fluid circulation from the bulk toward the surface along the domain boundary to push bubbles off and to convect them away from the surface. The goal is to avoid surface dryout, common in microgravity boiling, and to permit sustained, efficient surface cooling. If the calculations suggest this as being feasible, obvious microgravity/shuttle experiments would test it. In this and prior years we have extended our continuum calculation to describe the growth of individual bubbles on the surface and their detachment due to gravity, and developed a simple analytic theory to explain our findings. We have also begun to consider a constant heat flux, rather than a constant temperature heater condition and are examining the differences that it makes. We have completed a number of studies using LB and MD methods. See annual Task Book reports for details. In the last period and the future no cost extension year, we are examining the transition to film boiling, the situation where vapor takes over the entire solid surface momentarily, which could be catastrophic to sensitive electronics. We are using MD for this study. See Task Book extended abstract for details.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A vastly improved understanding of the mechanism of nucleate boiling heat transfer and the role played by gravity in it can have significant impacts on both terrestrial life and that on humans in space. An advance that makes nucleate boiling heat transfer essentially risk-free would enable the use of far less fluid to cool hot surfaces. For example, it could lead to the use of far less water in the secondary cooling of nuclear reactors, and therefore lead to less thermal pollution of our waters. Instead of releasing large amounts of hot water into rivers, one might be able to release far less hot water and more steam. Such an understanding might also influence computer design, since cooling issues are a critical constraint on computer design. In space applications, such cooling without the high risk of burnout could lead to far smaller cooling systems - and since the total space in the space station or in a space ship is extremely limited, such an advance could be highly beneficial.
	Progress Report: for NASA grant #NNX09AK40G. Title: How important is the contact line in nucleate boiling heat transfer? PI: David S. Rumschitzki, The City College of New York. Period: 16 April 2012-15 April 2013 Extended Abstract: This project is concerned with the development of a better understanding of how heat is transferred in nucleate pool boiling. In particular, it focuses on the three-phase (liquid-vapor-solid) contact line (CL) and its dynamics influence on the heat transfer in a system comprised of a vapor bubble in a liquid later on a solid that is heated from below. The work involves three approaches to this problem. 1. A continuum conduction-only model of quasi-steady heat transfer to a system with the given geometry, where one solves the quasi-steady heat equation in both the solid and the liquid regions, which are coupled to each other; 2. A molecular dynamics (MD) investigation of the nucleation, growth and detachment (the latter, in the case of the imposition of a uniform vertical body force) of a vapor bubble in a nanoscale version of the given geometry and heating; and 3. A lattice-Boltzmann (LB) method/calculation that considers the liquid and the vapor together a single fluid of non-uniform density, and its application to the growth and detachment (under gravity) of a vapor bubble from the solid surface in the given geometry and heating. We assume the reader has access to prior years' progress reports, and so we avoid repetition of their contents here. During the 2012-2013 year, we have focused on extending our work on the continuum and on the MD models and report on them below. We are currently writing up the LB results as well as those of the earlier continuum and MD results for publication. The earlier continuum results considered the case where, in the given geometry, the bottom surface of the solid, the one in contact with the heater and far from the liquid, was maintained at a constant temperature above the liquid's saturation temperature. Quasistatic calculations showed that the overwhelming majority of the heat from the heater made its way to the CL and that is where nearly all of the evaporation took place. Since the motion of a CL even without phase change is not well-understood, we first used ad hoc models (fixed contact angle or kinematic) to extend this calculation to follow the growth of a vapor bubble and found the volume of such bubbles generally grows as t3/2, t being time, until shortly before detachment, where the bubble deforms and the power decreases. MD offered the possibility of using the fundamental physical interactions

	between molecules in order to find macroscopic CL motion under bubble growth conditions and LB a mesoscopic method to follow bubble growth with CL motion an output that depended on system parameters and on the interaction parameter between the fluid and solid wall elements. This year, we extended the continuum calculation in two directions The first addresses the controversy in the literature as to whether the bulk of the heat transfer in nucleate boiling is via heat transfer around the vapor bubble or via evaporation into the bubble (with subsequent condensation at the top of the bubble and/or with bubble escape and rise). Using our continuum model we plot the fraction of the heat from the heater that goes into evaporation into the bubble and the fraction that goes around the bubble as a function of the system parameters. We give details below, but, in short, there are parameters where each of the two possible interpretations is realized.
Task Progress:	This year we have also considered the case where the heater, rather than supplying whatever local heat flux is necessary to maintain the bottom surface of the solid at fixed, uniform temperature, instead supplies a fixed, uniform heat flux. In this latter incarnation, we investigate how system parameters and the choice of CL model affect the (power law) growth of single bubbles. In the narrative, we detail these effects. Since the heater output is now uniform and constant in time and not of arbitrary local capacity, the scaling of bubble volume with time turns out to be different from (and slower than) that found in the uniform solid bottom temperature case. More interestingly, the power law dependence on time is no longer parameter- and CL-model-independent. We are carrying out numerous test cases in order to understand this behavior better. We are also trying to come up with a simpler analytical model to explain the resulting powers, as we had previously done for the uniform temperature heater case. This a major direction that we plan to follow in the 2013-2014 no-cost extension year. Our MD work has also progressed. Last year we followed bubbles as they nucleated, grew by evaporation and, given a uniform vertical body force, detached from the solid. We looked at the temperature distribution in the top solid layer and in the bottom fluid layer and superimposed the contact line on both of these. We had found that the temperature of the CL in the fluid essentially followed the locus of points in the lowest layer of the fluid that were roughly at the fluid saturation temperature. Our continuum model assumed that the CL was a continuation of the liquid-vapor interface and thus at the saturation temperature, and this result confirms that assumption. However, due to temperature slip (a lack of perfect continuity in temperature at the solid-fluid interface), the CL was not coincident with points on the top of the solid that were at the saturation temperature. We have used the CL evolutions predicted by these MD calculations in the continuum model to prescribe CL dynamics and calculated towards snapoff. We find the snapoff volume and the bubble reside volume and these values compare well with those predicted by running the MD to snapoff. The volume of the residue bubble after snapoff is critical, in our view, for the distinction between the “single bubble” and the “jets and columns” subregions of nucleate boiling: Small bubbles that are quenched by cold fluid coming to replace the rising detached bubble encourage the former, whereas large bubbles, having a CL and rapidly drawing heat from through the solid, encourage the latter. The other direction that we have taken our MD work is to try to understand the transition from nucleate to film boiling, In the former, there are both wet and dry spots that coexist on the solid surface, whereas in the latter case, the dry sots take over the entire solid. We have mapped out a set of temperatures (for the bottom layer of the solid) and pressures and followed our simulations long enough to determine for which cases the surface remains partially wet and for which vapor eventually takes over the entire surface. In those latter cases we double surface dimensions to check that this take-over is not a function of the size of the periodicity. In these early calculations, we choose pressures that are all zero or slightly negative, but where no bubble nucleates before we raise the temperature of the bottom layer of the solid and thermostat it. We are currently extending this to positive pressures, where we must use a higher solid bottom temperature and run the simulations much longer in order to clarify these questions. This is one of the major focuses of the work that we plan to continue into the 2012-2014 no-cost extension year.

Bibliography Type:	Description: (Last Updated: 04/09/2020)
Abstracts for Journals and Proceedings	Bao J, Koplik J, Rumschitzki DS. "Molecular dynamics simulation of vapor bubble growth and detachment on a heating surface." 2012 AIChE Annual Meeting, Pittsburgh, PA, October 28-November 2, 2012. Accepted for presentation at the meeting; postponed due to hurricane. 2012 AIChE Annual Meeting, Pittsburgh, PA, October 28-November 2, 2012. http://www3.aiche.org/Proceedings/Abstract.aspx?ConfID=Annual-2012&GroupID=1674&SessionID=22911&PaperID=281794 ; accessed 5/20/2013. , Nov-2012
Abstracts for Journals and Proceedings	Zheng L, Lee TH, Rumschitzki DS. "The growth of a vapor bubble on a heated surface using the lattice-Boltzmann calculation: Flow, heat transfer and phase change." 2012 AIChE Annual Meeting, Pittsburgh, PA, October 28-November 2, 2012. Accepted for presentation at the meeting; postponed due to hurricane. 2012 AIChE Annual Meeting, Pittsburgh, PA, October 28-November 2, 2012. http://www3.aiche.org/Proceedings/Abstract.aspx?ConfID=Annual-2012&GroupID=1674&SessionID=22912&PaperID=282964 ; accessed 5/20/13. , Nov-2012
Articles in Peer-reviewed Journals	Zheng L, Lee TH, Guo Z, Rumschitzki D. "Shrinkage of bubbles/drops in the lattice Boltzmann equation method for nonideal gases." Submitted to Physical Review E, June 2013. , Jun-2013